3.5.9

Solution: We start by finding the eigenvalues for A:

$$\det(A-\lambda I) = \begin{vmatrix} 5-\lambda & -3 & -2 \\ 8 & -5-\lambda & -4 \\ -4 & 3 & 3-\lambda \end{vmatrix} \xrightarrow{R_3\to R_1+R_3} \begin{vmatrix} 5-\lambda & -3 & -2 \\ 8 & -5-\lambda & -4 \\ 1-\lambda & 0 & 1-\lambda \end{vmatrix}$$
$$\xrightarrow{R_1\to 2R_1-R_2} \begin{vmatrix} 2-2\lambda & -3 & 0 \\ 8 & -5-\lambda & -4 \\ 1-\lambda & 0 & 1-\lambda \end{vmatrix}$$
$$\xrightarrow{C_1\to C_1+2C_3} \begin{vmatrix} 2-2\lambda & -3 & 0 \\ 0 & -5-\lambda & -4 \\ 3-3\lambda & 0 & 1-\lambda \end{vmatrix}$$
$$\Rightarrow (\lambda-1)^3 = 0 \Rightarrow \lambda_{1,2,3} = 1.$$

The associated eigenvector is

$$A - I = \begin{pmatrix} 4 & -3 & -2 \\ 8 & -6 & -4 \\ -4 & 3 & 2 \end{pmatrix} \to \begin{pmatrix} 4 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \vec{u} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Now since

$$(A-I)^2 = \mathbf{0}$$

every nonzero vector in $\mathbb{R}^3$ is a generalized eigenvector of eigenvalue 1. Then,

$$e^{tA} = e^t\left(I + t\begin{pmatrix} 4 & -3 & -2 \\ 8 & -6 & -4 \\ -4 & 3 & 2 \end{pmatrix}\right) = \boxed{e^t\begin{pmatrix} 1+4t & -3t & -2t \\ 8t & 1-6t & -4t \\ -4t & 3t & 1+2t \end{pmatrix}}$$

3.5.11:

Solution: We start by finding the eigenvalues of A:

$$\det(A-\lambda I) = \begin{vmatrix} 4-\lambda & -1 & 2 \\ 1 & 2-\lambda & 2 \\ -2 & 2 & 1-\lambda \end{vmatrix} \xrightarrow{R_2\to R_2-R_1} \begin{vmatrix} 4-\lambda & -1 & 2 \\ \lambda-3 & 3-\lambda & 0 \\ -2 & 2 & 1-\lambda \end{vmatrix}$$
$$\xrightarrow{C_2\to C_2+C_1} \begin{vmatrix} 4-\lambda & 3-\lambda & 2 \\ \lambda-3 & 0 & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix}$$
$$\Rightarrow (\lambda-3)^2(1-\lambda) \Rightarrow \lambda_{1,2} = 3 \text{ and } \lambda_3 = 1.$$

The corresponding eigenvectors are

$$A - 3I = \begin{pmatrix} 1 & -1 & 2 \\ 1 & -1 & 2 \\ -2 & 2 & -2 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \vec{u_1} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$
$$A - I = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 1 & 2 \\ -2 & 2 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 2 \\ 3 & -1 & 2 \\ -2 & 2 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 2 \\ 0 & -4 & -4 \\ 0 & 4 & 4 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 2 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \vec{u_3} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

We must also find a generalized eigenvector since $\lambda = 3$ has a multiplicity of 2 but the dimension of its eigenspace is $1 < 2$. We can find the generalized eigenvector by solving the equation

$$0 = (A - 3I)^2\vec{w} = \begin{pmatrix} -4 & 4 & -4 \\ -4 & 4 & -4 \\ 4 & -4 & 4 \end{pmatrix} \vec{w} \to \begin{pmatrix} -1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \vec{w} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Then,

$$P = \begin{pmatrix} \vec{u_1} & \vec{u_3} & \vec{w} \end{pmatrix} = \begin{pmatrix} -1 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

so the Jordan form is

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

For the 2×2 Jordan block $J_2(3) = 3I + N$ where

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Also

$$e^{tJ_2(3)} = e^{t(3I+N)} = e^{3t}e^{tN} = e^{3t}(I + tN) = e^{3t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix},$$

so

$$e^{tJ} = \begin{pmatrix} e^t & 0 & 0 \\ 0 & e^{3t} & te^{3t} \\ 0 & 0 & e^{3t} \end{pmatrix}.$$

Since $e^{tA} = Pe^{tJ}P^{-1}$, we have

$$\boxed{e^{tA} = e^{3t}\begin{pmatrix} 2-t & t-1 & 1 \\ 1-t & t & 1 \\ -1 & 1 & 0 \end{pmatrix} + e^t \begin{pmatrix} -1 & 1 & -1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}}$$

4.5.2:

Solution: We start by finding e^{tA}. The eigenvalues of A are given by the characteristic polynomial so

$$\lambda^2 - 10\lambda + 16 \Rightarrow (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 8, \lambda_2 = 2.$$

The associated eigenvectors are

$$A - 8I = \begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix} \to \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow \vec{u_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$A - 2I = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow \vec{u_2} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Then,

$$V = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \Rightarrow V^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \Rightarrow e^{tA} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} e^{8t} & 0 \\ 0 & e^{2t} \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix} e^{8t}+e^{2t} & e^{8t}-e^{2t} \\ e^{8t}-e^{2t} & e^{8t}+e^{2t} \end{pmatrix}\vec{x}.$$

So,

$$\vec{x}'(t) = A\vec{x}(t) + \vec{g}(t) \Rightarrow (e^{-tA}\vec{x})' = e^{-tA}\vec{g} \Rightarrow e^{-tA}\vec{x} - e^{-t_0A}\vec{x}_0 = \int_{t_0}^{t} e^{sA}\vec{g}(s)\,\mathrm{d}s$$
$$\Rightarrow \vec{x}(t) = e^{t-t_0}\vec{x_0} + \int_{t_0}^{t} e^{(t-s)A}\vec{g}(s)\,\mathrm{d}s.$$

But,

$$\vec{g}(s) = e^{-8s}(s,s) = se^{-8s}(1,1)$$

and

$$e^{(t-s)A}(1,1) = e^{8(t-s)}(1,1)$$

since 8 is an eigenvalue of A. So

$$\int_0^t e^{(t-s)A}e^{-8s}(s,s)^T\,\mathrm{d}s = \int_0^t se^{-8s}e^{8(t-s)}\,\mathrm{d}s\,(1,1)^T$$
$$= e^{8t}\left(\int_0^t se^{-16s}\,\mathrm{d}s\right)\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$= e^{8t}\left(\frac{1}{256} - e^{-16t}\left(\frac{t}{16}+\frac{1}{256}\right)\right)\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$= \frac{e^{8t}-(16t+1)e^{-8t}}{256}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Since

$$e^{tA}\vec{x}_0 = \begin{pmatrix} \frac{e^{8t}-e^{2t}}{2} \\ \frac{e^{8t}+e^{2t}}{2} \end{pmatrix}$$

we have

$$\boxed{\vec{x}(t) = \begin{pmatrix} \frac{e^{8t}-e^{2t}}{2} \\ \frac{e^{8t}+e^{2t}}{2} \end{pmatrix} + \frac{e^{8t}-(16t+1)e^{-8t}}{256}\begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

4.5.8:

Solution:
(a) We must solve the system $\vec{v}(x,y) = \vec{0}$ to the equilibrium points. This is equivalent to

$$0 = (x+y)(x-y-1) \Rightarrow x+y=0 \quad \text{or} \quad x-y-1=0$$
$$0 = (x+y-2)(x-y+1) \Rightarrow x+y-2=0 \quad \text{or} \quad x-y+1=0.$$

The equilibrium points will lie at the intersection between pairs of these equations. One can see that the only consistent pairs are

$$x + y = 0 \quad \text{and} \quad x - y + 1 = 0 \Rightarrow (x, y) = \left(-\frac{1}{2}, \frac{1}{2}\right)$$
$$x - y - 1 = 0 \quad \text{and} \quad x + y - 2 = 0 \Rightarrow (x, y) = \left(\frac{3}{2}, \frac{1}{2}\right).$$

Linearizing we see that

$$f(x, y) = (x + y)(x - y - 1) \quad \text{and} \quad g(x, y) = (x + y - 2)(x - y + 1)$$

so

$$D_v(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} 2x - 1 & -2y - 1 \\ 2x - 1 & -2y + 3 \end{pmatrix}.$$

Then

$$D_v\left(-\frac{1}{2}, \frac{1}{2}\right) = \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix} \Rightarrow \det(D_v) = -8 < 0$$
$$D_v\left(\frac{3}{2}, \frac{1}{2}\right) = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix} \Rightarrow \lambda_{1,2} = 2 \pm 2i.$$

It follows that **both equilibrium points are unstable.**

(b) We know from part (a) that the equilibrium points lie at

$$\left(-\frac{1}{2}, \frac{1}{2}\right) \quad \text{and} \quad \left(\frac{3}{2}, \frac{1}{2}\right).$$

Then,

$$f(x, y) = (x + y - 2)(x - y + 1) \quad \text{and} \quad g(x, y) = (x + y)(x - y - 1)$$

so

$$D_v(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} 2x - 1 & -2y + 3 \\ 2x - 1 & -2y - 1 \end{pmatrix}.$$

Then

$$D_v\left(-\frac{1}{2}, \frac{1}{2}\right) = \begin{pmatrix} -2 & 2 \\ -2 & -2 \end{pmatrix} \Rightarrow \lambda_{1,2} = -2 \pm 2i$$
$$D_v\left(\frac{3}{2}, \frac{1}{2}\right) = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} \Rightarrow \det(D_v) = -8 < 0$$

It follows that $\left(-\frac{1}{2}, \frac{1}{2}\right)$ **is stable** and $\left(\frac{3}{2}, \frac{1}{2}\right)$ **is unstable**.

4.5.9:

Solution:

(a) Just like in (4.5.8) we start by solving $\vec{v}(x, y) = \vec{0}$. This is equivalent to

$$x - xy = x(1 - y) = 0 \Rightarrow x = 0 \quad \text{or} \quad y = 1$$
$$y + 2xy = y(1 + 2x) \Rightarrow x = -\frac{1}{2} \quad \text{or} \quad y = 0.$$

So the equilibrium points are $(0,0)$ and $\left(-\frac{1}{2},1\right)$. Then,

$$f(x,y) = x - xy \quad \text{and} \quad g(x,y) = y + 2xy$$

so

$$D_v(x,y) = \begin{pmatrix} 1-y & -x \\ 2y & 1+2x \end{pmatrix}.$$

Thus,

$$D_v(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \lambda_{1,2} = 1$$

$$D_v\left(-\frac{1}{2},1\right) = \begin{pmatrix} 0 & \frac{1}{2} \\ 2 & 0 \end{pmatrix} \Rightarrow \det(D_v) = -1 < 0.$$

It follows that **both equilibrium points are unstable**.

(b) From (a) we know that $(0,0)$ and $\left(-\frac{1}{2},1\right)$ are equilibrium points. Then

$$f(x,y) = y + 2xy \quad \text{and} \quad g(x,y) = x - xy$$

so

$$D_v(x,y) = \begin{pmatrix} 2y & 1+2x \\ 1-y & -x \end{pmatrix}.$$

Thus,

$$D_v(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow \det(D_v) = -1 < 0$$

$$D_v\left(-\frac{1}{2},1\right) = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \Rightarrow \lambda_1 = 2, \lambda_2 = \frac{1}{2} \Rightarrow \Re(\lambda_1) > 0, \Re(\lambda_2) > 0.$$

It follows that **both equilibrium points are unstable**.